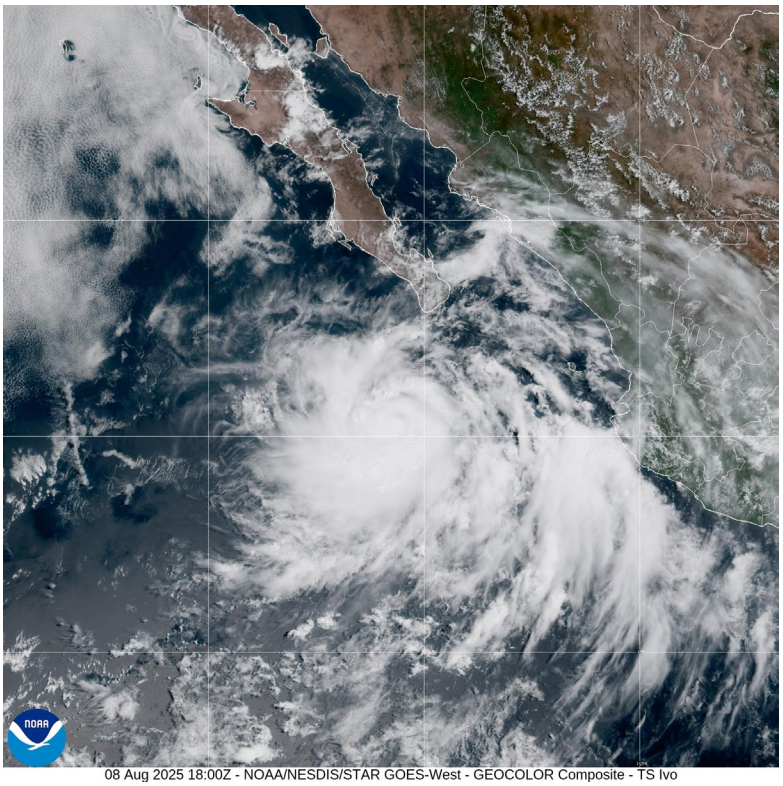
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

TROPICAL STORM IVO

(EP092025)

6–11 August 2025

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National Hurricane Center
23 October 2025



GOES-18 GEOCOLOR IMAGE OF IVO AT 1800 UTC 08 AUGUST. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Ivo was a small tropical storm that passed just offshore of the southwestern coast of Mexico.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Ivo.

Tropical Storm Ivo

6–11 AUGUST 2025

BEST TRACK

The "best track[2]" positions and intensities for Tropical Storm Ivo are listed in Table 1. The best track chart of Ivo's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively.

There were no reliable reports of tropical-storm force winds associated with Ivo.

Origin

Ivo formed from a low-pressure area that developed along the monsoon trough southwest of Costa Rica on 4 August, a development likely related to a tropical wave that crossed Panama on 3 August. The system moved west-northwestward parallel to the coast of Central America and southeastern Mexico with increasing organization, and a tropical depression developed near 1200 UTC 6 August about 295 n mi southeast of Acapulco, Mexico.

Peak Intensity and Minimum Pressure

Ivo's peak intensity of 55 kt at 1800 UTC 8 August and 0000 UTC 9 August is based on a blend of subjective and objective Dvorak Technique intensity estimates. It should be noted in Fig. 2 that two scatterometer overpasses occurred near the time of the peak intensity that indicated lower winds. However, the scatterometer was unlikely to have resolved the maximum winds in Ivo's small inner core, which was apparent in a microwave image at 1600 UTC 8 August (Fig. 4).

The estimated minimum central pressure of 998 mb is based on the Knaff-Zehr-Courtney wind-pressure relationship for Ivo's peak intensity and small size at that time.

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[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Ivo.

CASUALTY AND DAMAGE STATISTICS

There were no reports of casualties associated with Ivo. The Meteorological Service of Mexico reports that Ivo produced rains of 1–3 inches along portions of the southwestern coast of Mexico with local amounts around 5 inches. Media reports indicate that these rains caused minor damage from freshwater flooding and mudslides.

FORECAST AND WARNING VERIFICATION

Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. The formation of Ivo was generally well forecast, with the system being introduced into the 7-day TWO 126 h before genesis and the 2-day TWO 54 h before genesis. Figure 5 shows composites of 7-day TWO genesis areas for each category (low <40%, medium 40-60%, and high >60%) prior to the formation of Ivo. The spatial forecast generally caught the genesis location well, although the formation location was near the northeastern edge of the genesis forecast areas.

A verification of NHC official track forecasts for Ivo is given in Table 3a. Official track forecast errors were greater than the mean official errors for the previous 5-yr period for the forecast times from 12–72 h, and less than the 5-yr mean at 96 h. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. Many of the track guidance models had lower average errors than the official forecast, with two of the better performers being the regional hurricane models HAFS-A (HFAI) and COAMPS-TC (CTCI). Another good performer was the experimental Google DeepMind AI model (GDMI), which had lower mean errors than the official forecast at all forecast periods. Examination of the individual official forecasts (not shown) shows that they followed the general motion of the storm well. However, several of the forecasts had larger errors during a period when Ivo's low-level center was difficult to track and the estimated operational positions differed from the best track positions.

A verification of NHC official intensity forecasts for Ivo is given in Table 4a. Official intensity forecast errors were lower than the mean official errors for the previous 5-yr period. However, the official forecasts had larger average errors than Climatology-Persistence (OCD5) at the 24–60 h forecast times, indicating these forecasts had no skill. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. Several of the intensity guidance models had lower average errors than the official forecasts at most forecast times, including the intensity consensus (IVCN), the HFIP Correct Consensus (HCCA), and the Florida State Superensemble (FSSE). Examination of the individual official forecasts (not shown) indicates they had a consistent high bias, with Ivo not becoming as strong as forecast.

Although Ivo was close to the coast of Mexico, no coastal watches or warnings were issued due to the small size of the cyclone.

ACKNOWLEDGEMENTS

Philippe Papin (NHC) created the Graphical Tropical Weather Outlook verification figure.

Table 1. Best track for Tropical Storm Ivo, 6–11 August 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
06 / 1200	13.0	96.5	1009	30	tropical depression
06 / 1800	13.8	98.2	1008	35	tropical storm
07 / 0000	14.7	99.9	1007	35	"
07 / 0600	15.8	101.5	1006	40	"
07 / 1200	17.2	103.2	1004	45	"
07 / 1800	18.3	105.0	1001	50	"
08 / 0000	19.3	106.6	1001	50	"
08 / 0600	20.0	108.1	1001	50	"
08 / 1200	20.2	109.5	1000	50	"
08 / 1800	20.3	110.7	999	55	"
09 / 0000	20.5	111.3	998	55	"
09 / 0600	20.7	111.9	999	50	"
09 / 1200	20.8	112.5	1001	50	"
09 / 1800	20.9	113.1	1002	45	"
10 / 0000	21.1	113.7	1003	45	"
10 / 0600	21.3	114.5	1004	40	"
10 / 1200	21.6	115.6	1005	35	"
10 / 1800	22.0	116.6	1006	35	"
11 / 0000	22.3	117.4	1007	30	tropical depression
11 / 0600	22.6	118.2	1008	30	remnant low
11 / 1200	23.0	119.1	1009	25	"
11 / 1800	23.3	120.0	1010	20	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
12 / 0000	23.5	121.0	1010	20	"
12 / 0600					dissipated
09 / 0000	20.5	111.3	998	55	maximum winds and minimum pressure

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	54	126
Medium (40%-60%)	30	102
High (>60%)	18	42

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Tropical Storm Ivo, 6–11 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	30.5	47.9	55.9	66.5	76.5	79.9	**63.4**	
OCD5	49.0	104.8	140.2	168.7	170.3	136.0	94.5	
Forecasts	16	14	12	10	8	6	2	
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Tropical Storm Ivo, 6–11 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	30.3	46.1	54.7	71.2	81.1	74.5	63.5	
OCD5	49.4	101.8	134.2	166.7	150.0	138.6	79.8	
GFSI	**25.1**	**36.4**	**45.5**	74.9	115.4	112.5	126.7	
HWFI	38.7	59.1	77.1	106.1	166.1	187.4	223.5	
HMNI	32.7	47.2	59.6	79.7	88.2	98.8	**58.5**	
HFAI	**27.5**	**39.7**	**40.7**	**57.1**	**66.8**	**68.0**	96.3	
HFBI	32.9	49.2	**50.3**	**67.2**	85.5	79.8	**45.5**	
EMXI	34.0	54.4	67.0	87.5	84.3	110.8	131.1	
NVGI	35.4	54.7	**53.6**	**61.1**	**61.1**	87.0	**30.3**	
CMCI	**30.2**	**43.6**	**44.1**	**60.4**	**62.7**	94.5	75.4	
CTCI	**24.6**	**37.7**	**38.4**	**54.4**	**62.7**	**70.4**	247.2	
TVCE	**28.6**	**40.5**	**44.5**	**63.8**	81.4	81.0	71.6	
TVCX	**28.8**	**40.1**	**45.9**	**64.6**	83.4	84.4	74.7	
GFEX	**29.0**	**40.6**	**52.5**	79.0	100.4	110.0	126.8	
TVDG	**29.3**	**41.0**	**47.4**	**66.7**	86.3	85.9	82.9	
HCCA	31.4	**45.8**	**53.5**	74.2	86.3	76.6	**21.2**	
FSSE	**29.4**	**41.5**	**47.3**	71.6	87.0	84.3	**45.7**	
AEMI	**28.1**	**40.8**	**48.4**	76.6	107.2	108.0	**53.7**	
GDMI	**27.6**	**40.5**	**45.9**	**52.9**	**56.8**	**53.3**	**56.5**	
TABS	39.2	65.7	79.5	101.9	153.3	168.0	211.8	
TABM	37.1	57.9	76.1	102.5	119.6	141.0	98.9	
TABD	38.9	65.3	78.6	97.5	96.6	105.1	96.0	
Forecasts	15	13	11	9	5	5	1	

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Tropical Storm Ivo, 6–11 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**3.4**	**7.9**	**10.4**	**10.5**	**7.5**	**6.7**	**7.5**	
OCD5	4.6	5.4	8.2	6.8	6.8	8.0	22.0	
Forecasts	16	14	12	10	8	6	2	
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Tropical Storm Ivo, 6–11 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	3.7	8.5	10.5	10.0	7.0	5.0	10.0	
OCD5	4.6	**5.2**	**7.9**	**6.6**	**6.0**	7.2	24.0	
HWFI	5.6	**8.0**	11.8	10.8	11.0	12.6	36.0	
HMNI	6.5	12.3	17.2	20.8	23.4	16.4	**3.0**	
HFAI	5.3	**7.5**	**9.4**	12.7	14.0	13.0	12.0	
HFBI	3.7	**8.0**	**8.8**	10.0	7.8	6.6	**8.0**	
DSHP	4.6	9.1	15.5	21.1	24.6	23.0	18.0	
LGEM	5.3	**6.4**	**10.4**	12.9	15.2	17.0	17.0	
ICON	4.7	**4.2**	**6.6**	**7.2**	**5.8**	8.2	17.0	
IVCN	3.9	**4.2**	**5.7**	**7.1**	**4.4**	**2.4**	**4.0**	
IVDR	3.9	**5.0**	**6.8**	**8.4**	7.0	**4.4**	**2.0**	
CTCI	4.8	**7.5**	12.2	16.6	24.6	22.2	19.0	
GFSI	5.0	**7.6**	**9.1**	**9.4**	7.4	7.0	**7.0**	
EMXI	5.5	9.8	14.5	16.3	19.4	18.6	**9.0**	
HCCA	4.3	**5.0**	**6.6**	**7.6**	**6.0**	9.6	**6.0**	
FSSE	**3.3**	**3.4**	**6.5**	**7.9**	10.8	9.4	**5.0**	
GDMI	5.5	**4.6**	**5.2**	**5.3**	**3.8**	**2.8**	12.0	
Forecasts	15	13	11	9	5	5	1	

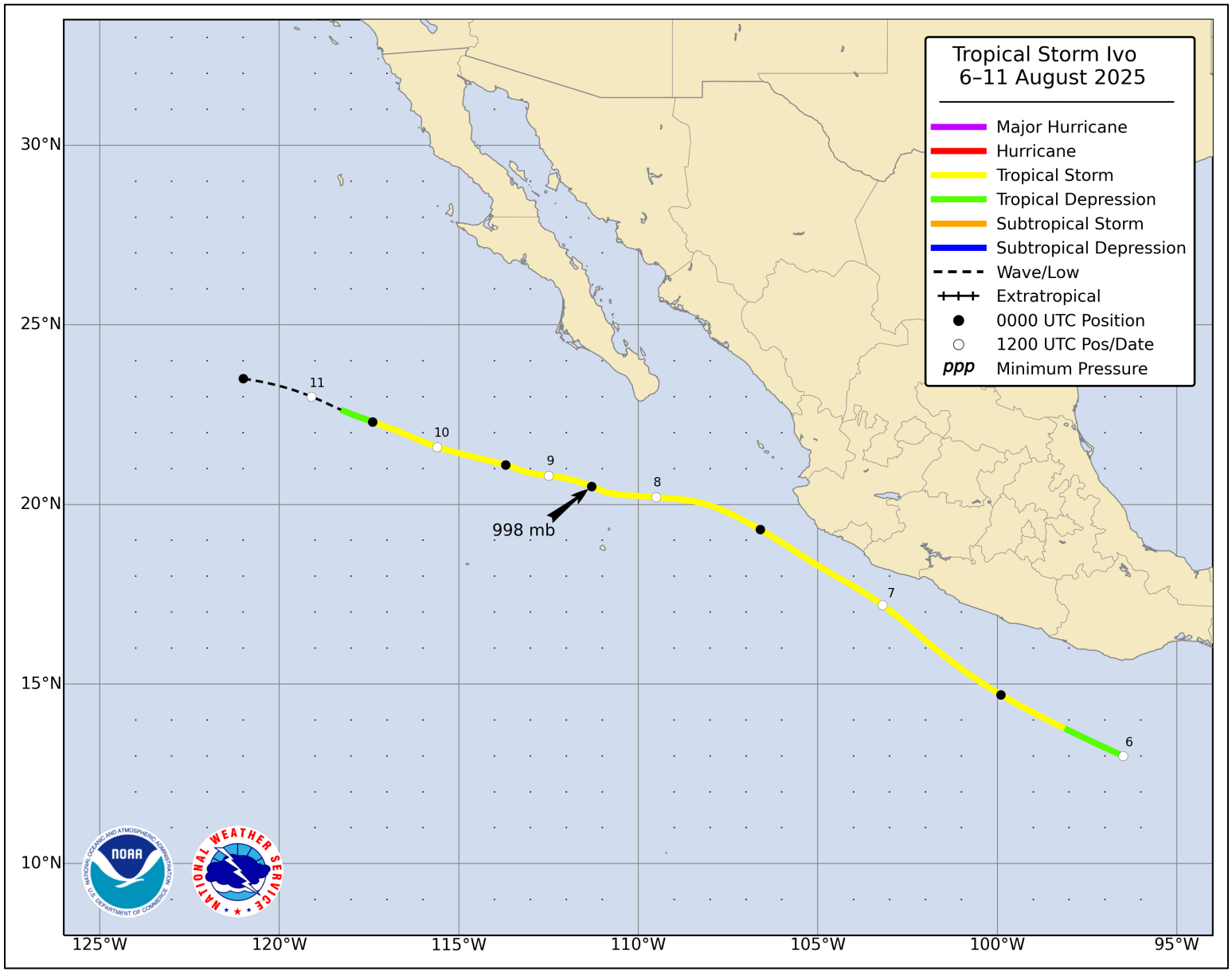


Figure 1. Best track positions for Tropical Storm Ivo, 6–11 August 2025.

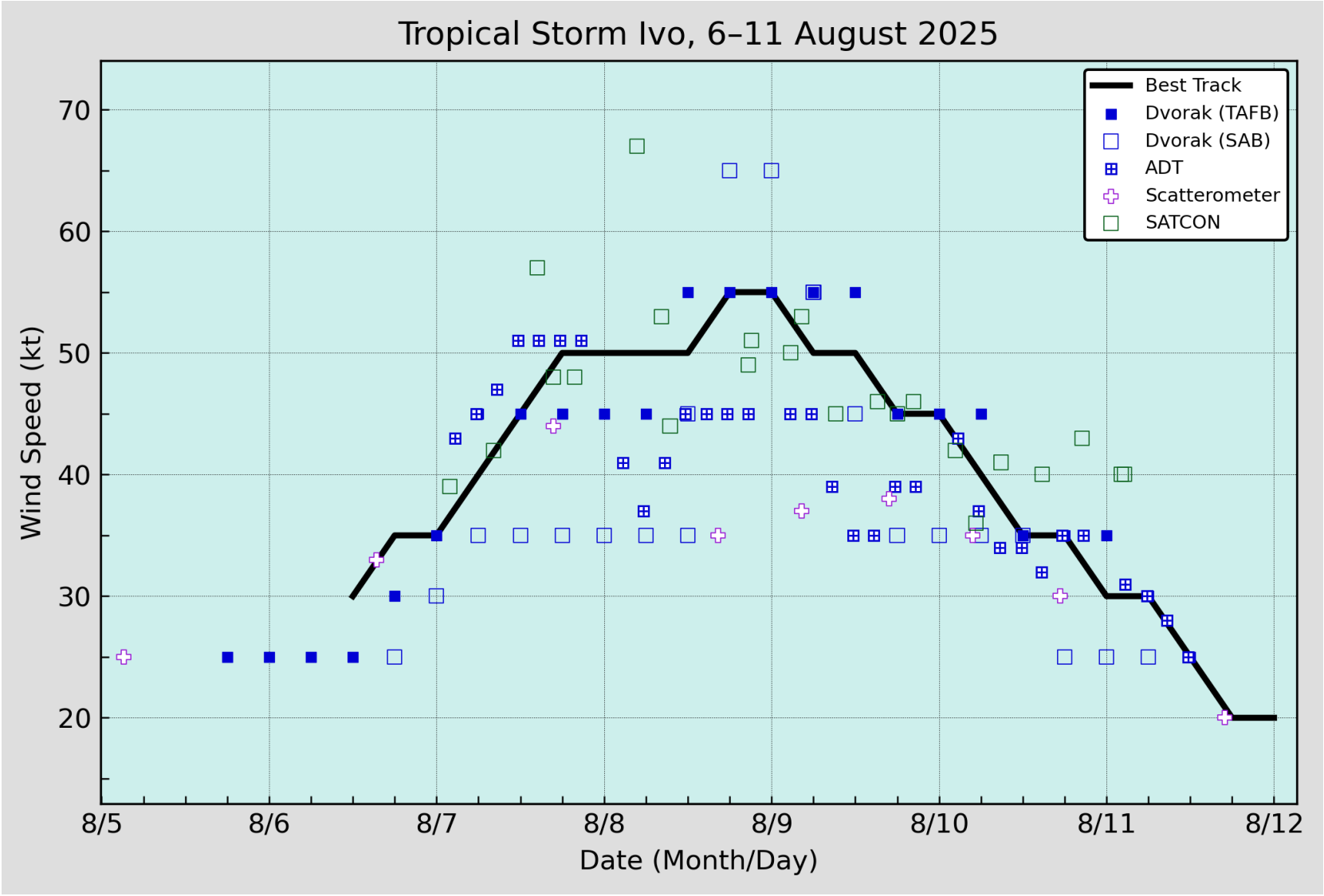


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Tropical Storm Ivo, 6–11 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

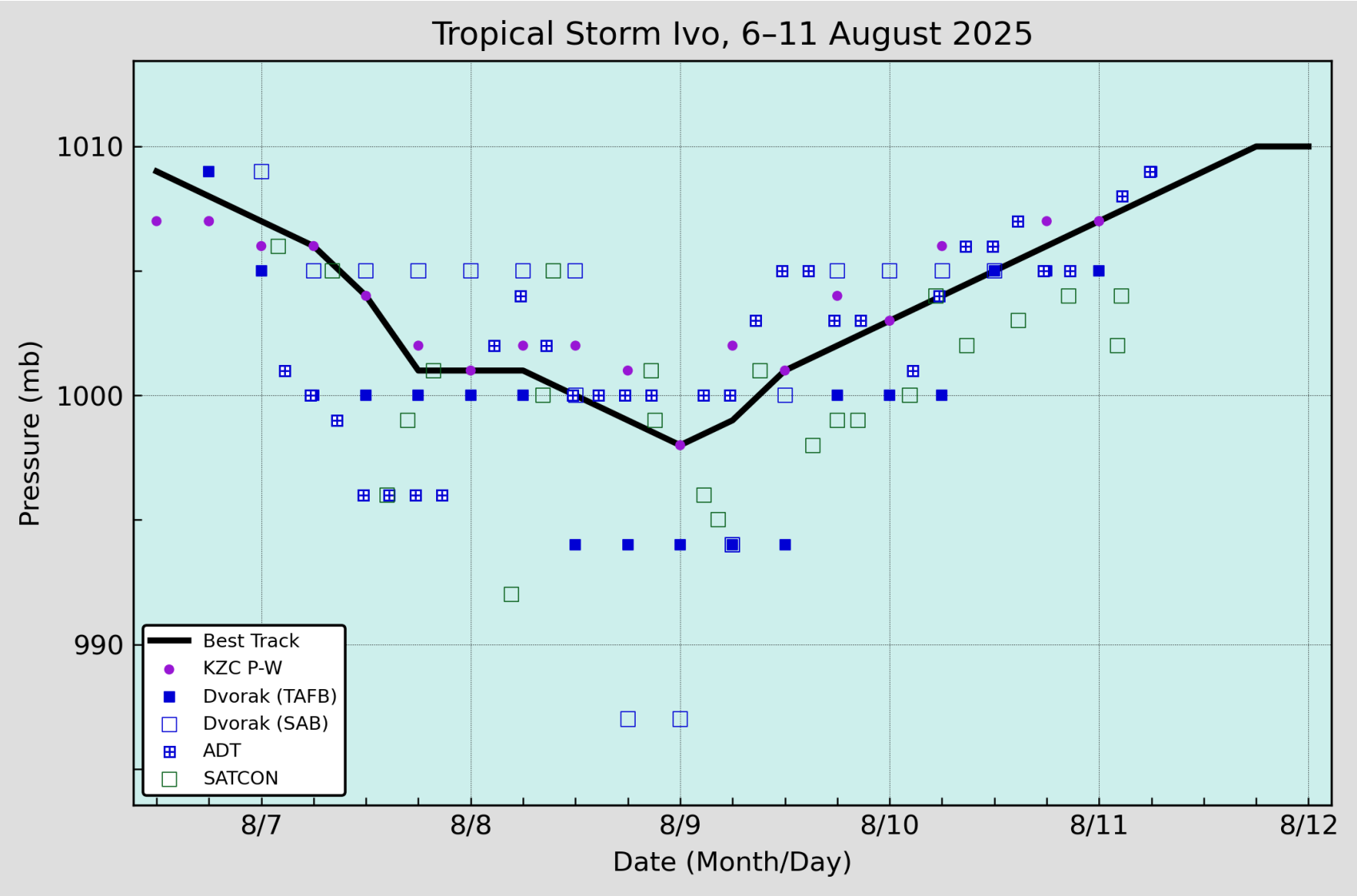


Figure 3. Selected pressure observations and best track minimum central pressure curve for Tropical Storm Ivo, 6–11 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

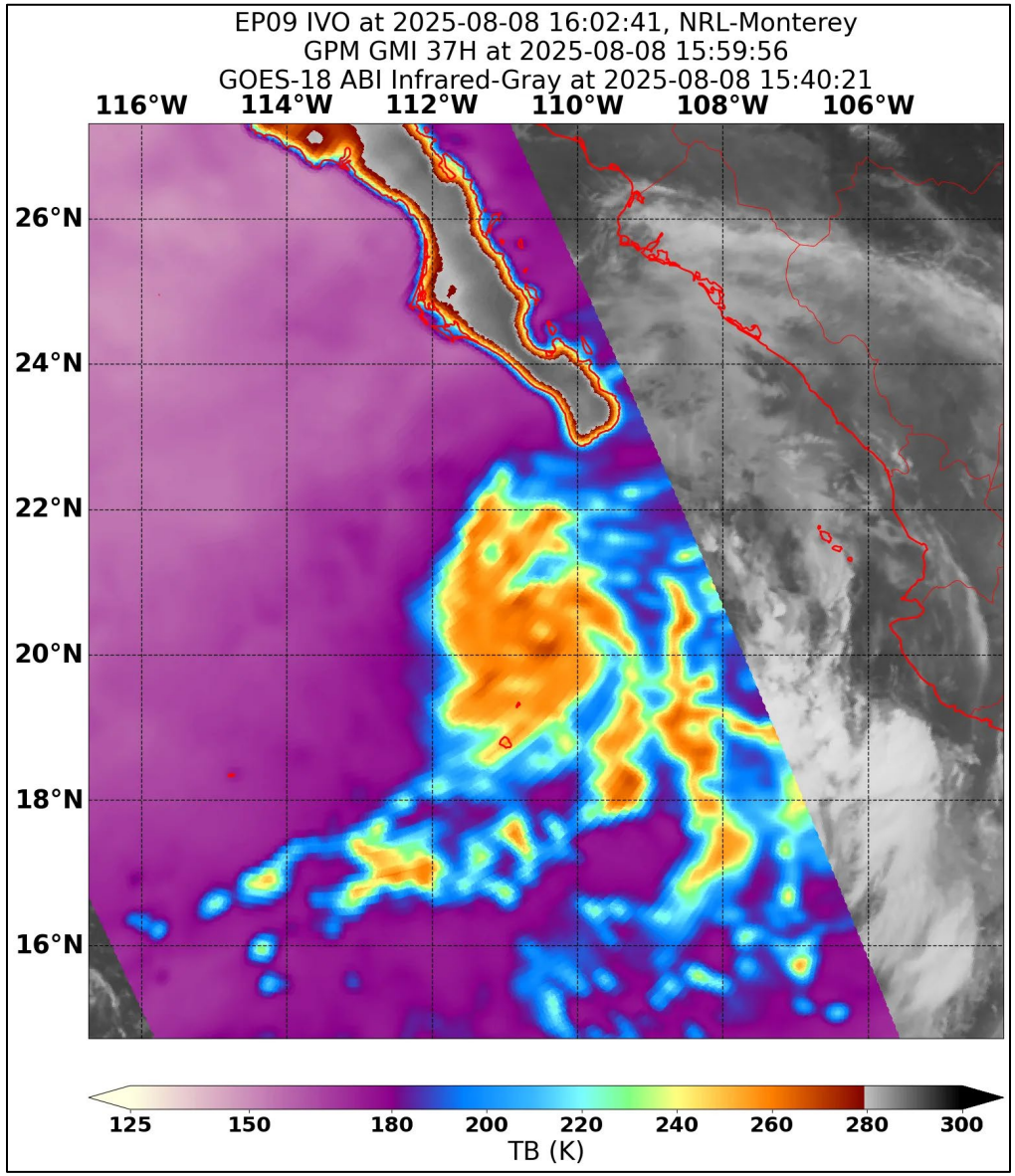


Figure 4. 37 GHz horizontal polarization image of Ivo at 1600 UTC 8 October from the NASA/JAXA GPM satellite. Image courtesy of the Naval Research Laboratory, Monterey, CA.

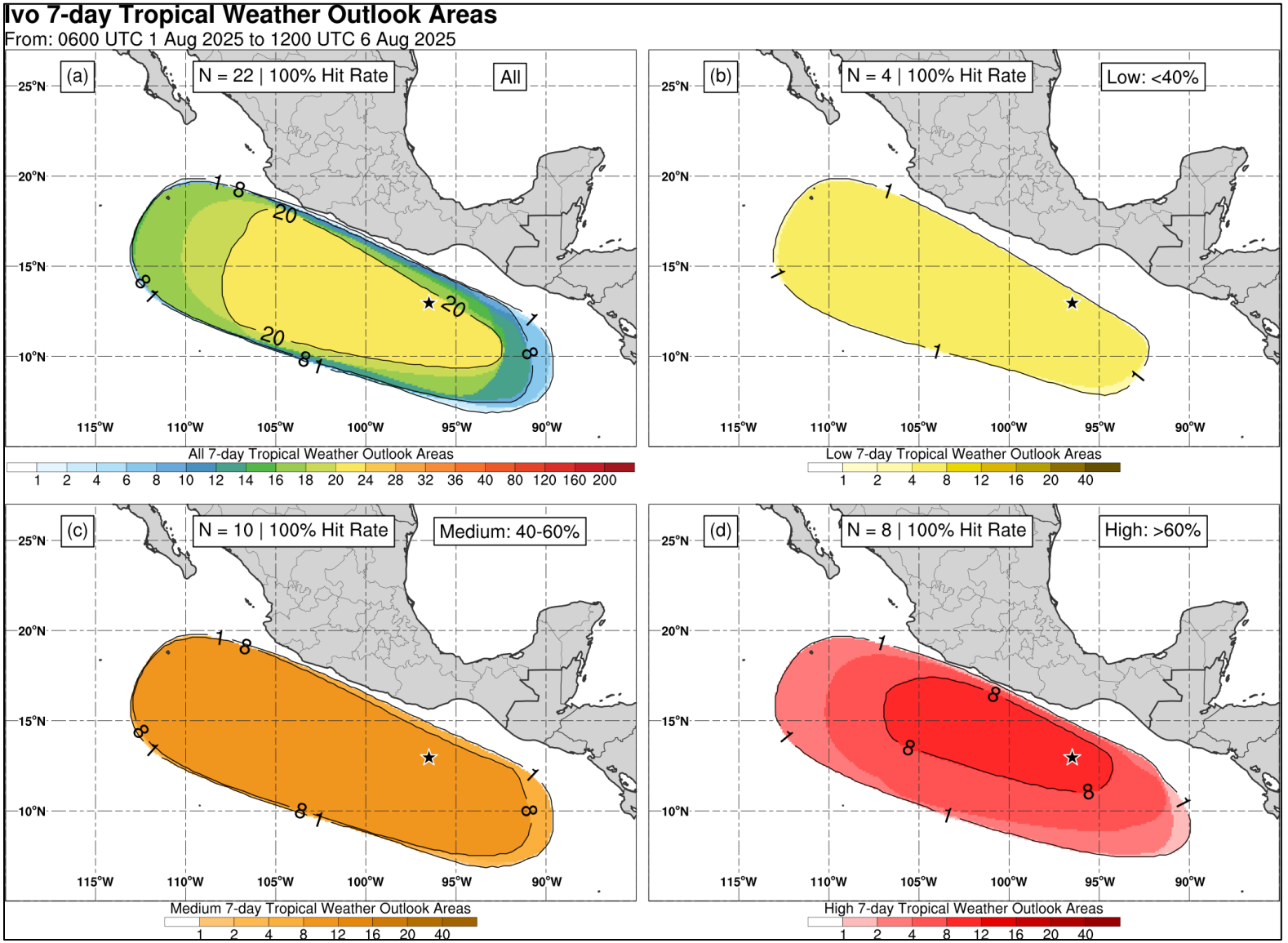


Figure 5. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Ivo for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.